



NUTRITIONAL ASSESSMENT (HuNu3081)

CHAPTER FOUR

BIOCHEMICAL/ LABORATORY ASSESSMENT METHODS

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Objectives

- **At the end of this chapter will be able to:**
 - ✓ Identify biochemical tests used as indicators of nutrient status.
 - ✓ Describe the purposes of biochemical tests.
 - ✓ Discuss the pros and cons of biochemical tests.
 - ✓ Identify tests employed in the assessment of protein.
 - ✓ Identify tests employed in the assessment of micronutrients (Iron, Vitamin A, Zinc, Iodine, Vitamin 12 & Folic Acid).

Introduction

- Laboratory tests are primarily used to:
 - ✓ detect sub-clinical deficiency states or
 - ✓ confirm nutritional status findings uncovered by other assessment methods.
- It is the most **objective** & **quantitative** method of nutritional assessment.
- However, a combination of laboratory tests should be used rather than a single test for each nutrient.
- To perform a laboratory test, a sample such as blood or urine must be collected and analyzed.

Purpose of Biochemical/ Laboratory Tests

- To recognize acute malnutrition for which the clinical signs are non-specific, *e.g.* potassium deficiency.
- To confirm the clinical diagnosis of a deficiency diseases, *e.g.* xerophthalmia, beriberi, scurvy, rickets, kwashiorkor etc.
- For monitoring nutritional managements in intensive care.
- To detect sub-clinical micronutrient deficiencies in community nutrition surveys.
- To demonstrate the response to a nutrition education program, *e.g.* reduction of plasma cholesterol.
- To diagnose nutritional supplement overdosing, *e.g.* Vitamin A.

Pros and Cons of Biochemical Tests

Pros

- ✓ Detect sub-clinical deficiency state, early changes in body metabolism & nutrition status (before the appearance of clinical signs).
- ✓ More objective (free of bias), precise, accurate & reproducible
- ✓ Useful to validate data obtained from dietary methods *e.g. comparing salt intake with 24-hour urinary excretion.*
- ✓ Give gradable results

Cons

- ✓ Not accessible for routine use
- ✓ Can't be applied on large scale
- ✓ Many quality control problems
- ✓ No ideal site for sampling
- ✓ Need sophisticated instrument & highly trained staffs
- ✓ Involve invasive procedures
- ✓ Time consuming & expensive

Types of Biochemical/ Laboratory Tests

1. Static Biochemical Tests:

- Measure the level of nutrients in biological fluids and tissues or the urinary excretion rate of the nutrient or its metabolite.
 - ✓ It reflects the total body nutrient content or the nutrient store in tissue most sensitive to depletion.
- Mostly **whole blood** or some fraction of blood (**serum, erythrocytes & leucocytes**) used to measure a nutrient in a biological fluid or tissue.
 - ✓ *E.g.* Selenium levels can be measured in whole blood, folate in red blood cells, magnesium & vitamin C in leucocytes.
- Some of these static biochemical tests monitor short-term or acute changes in nutritional status (*e.g.* urinary excretion rate of a nutrient or its metabolite);
- Others provide a long-term or chronic index of nutrient status (*e.g.* Erythrocytes, concentration trace elements in hair, fingernails, & toenails).

Types of Biochemical/ Laboratory Tests... (1)

2. Functional Tests:

- Measure the extent of the functional consequences of a specific nutrient deficiency.
- They are diagnostic tests used to determine the sufficiency of host nutrient to permit cells, tissues, organs, enzymes to perform optimally the intended nutrient dependent biological function.
- They have greater biological significance than the static laboratory tests.
- They are **not suitable** for field surveys as they are **too invasive**.

2.1 Functional Biochemical Tests

- Functional Biochemical Tests can be:
 - **Tolerance tests/ plasma appearance tests**, (*conducted in vivo*);
E.g. Glucose tolerance test.
 - **Load tests**: used to assess deficiencies of water soluble vitamins (*conducted in vivo*).
E.g. Tryptophan load tests for pyridoxine (Vitamin B6), Histidine load test for folic acid, and Vitamin C load test.
- These tests assess the consequences of a nutrient deficiency by:
 - **Changes/ reduction in enzyme activities in the blood**
 - **Detection of abnormal metabolic products in urine or blood**:
(due to reduced activity of a nutrient-dependent enzyme).
E.g. Urinary creatinine
 - **Changes in concentration of specific blood components**:
E.g. Hemoglobin concentration in whole blood.

2.2 Functional Physiological Tests

- Assesses the physiological performance of an individual *in vivo*.
- It include tests of:
 - ✓ Immune competence,
 - ✓ Taste acuity (impaired in mild zinc deficiency),
 - ✓ Night blindness: assessed using a rapid dark adaptation (RDA) test,
 - ✓ Muscle function test (when protein stores are reduced & during protein energy malnutrition),
 - ✓ Growth and developmental responses such as sexual maturation, and cognitive function.
- None of these tests are specific and hence must be used in conjunction with biochemical tests.

Assessment of Protein Status

- Serum albumin and prealbumin are useful indicators of body protein status.
- However, care should be taken in interpreting results; because:
 - ✓ Tests are non-specific and non-sensitive,
 - ✓ Levels of protein may decrease in response to injury, disease, or inflammation, which means that individuals with low serum albumin & prealbumin levels may or may not be malnourished.
- **Serum albumin** measurement: an indicator of protein status over a period of weeks to months (useful for **severe malnutrition**).
 - Its value can be affected if:
 - ✓ **Protein losing enteropathy** occurs, (renal losses, or burns) and **liver disease** (decrease in rate of albumin synthesis).
 - ✓ Not sensitive or specific indicator of acute PEM.

Assessment of Protein Status ...(1)

- **Prealbumin (Transthyretin)** and **Retinol Binding Protein (RBP)**: useful indices of **acute protein depletion** but affected by **vitamin A deficiency (VAD)**.
- **Transferrin** (for **chronic deficiency**), not highly useful measure of protein status (can be increased by **infection & iron deficiency**).
- **Plasma amino acid concentrations**: change with acute alterations in the level of protein and amino acid intake (currently not recommended for routine use).
- **Fibronectin**: reduced in acute nutritional deprivation and returns to normal on early nutritional rehabilitation.

Assessment of Protein Status ...(2)

- **Measurements on Urine:** creatinine/height index (CHI).

$$\text{CHI} = \frac{\text{24-h urine creatinine excretion (mg)} \times 100}{\text{24-h creatine excretion (mg) by a normal subject of same height}}$$

- CHI 60-80%: mild protein depletion and
 - Less than 40%: evidence of **severe protein depletion**.
-
- **Functional Tests:** muscle function (hand-grip strength), immune system abnormalities and protein-energy malnutrition.

Assessment of Iron Status

- Iron is one of the essential trace elements for life.
- Iron is stored primarily as **ferritin** and **hemosiderin** (maintain iron homeostasis).

Sources: iron occurs in two forms.

- Heme iron:** obtained from animal products: meat, poultry & fish (MPF). *e.g.* **Hemoglobin** & **myoglobin**.
- Non-Heme iron:** obtained from **plant products, enriched products & dairy foods**.

Physiological functions:

- Binding and transport of oxygen, regulation of gene, cell growth & differentiation, immune & cognitive function, energy metabolism.

Recommendations

	Age	Iron (mg/day)
Infants	First 6 months	0.27
	7 – 12 months	11
Children	1 – 3 years	7
	4 – 8 years	10
Teen age boys	9 – 13 years	8
	14 – 18 years	11
Teen age girls	9 – 13 years	8
	14 – 18 years	15
Adult men	>= 19 years	8
Adult women	19 – 50 years	18
Adults	>= 51 years	8
Pregnant women		27
Lactating women	< 18 years	10
Lactating women	19 – 50 years	9



- **Causes of Iron deficiency?**
- **Consequences of Iron Deficiency?**
- **Who are at risk of Iron deficiency?**

Causes of IDA

▪ Dietary:

- ✓ Inadequate dietary intakes; poor absorption (plant-based diets: high anti-nutritional factors *i.e.* low bioavailability), deficiencies of vitamin A, B12, folic acid & possibly B6.

▪ Non-dietary

- ✓ Excessive losses: menstruation; frequent blood donations; bleeds; malaria, intestinal parasites; haemolysis.
- ✓ High physiological requirements, infancy, adolescence, pregnancy & lactation
- ✓ Inflammation caused by infection

Consequences of Iron Deficiency:

- ✓ Anemia (IDA), Poor growth & cognitive development, Poor resistance to infection, Poor work capacity *i.e.* poor productivity, Impair academic achievement: IDA lowers IQ by 10 points.

At Risk Groups for iron deficiency

- IYC: low iron in milk, rapid growth & low iron stores
- Adolescents in growth spurt
- Females during childbearing years
- Pregnant women: expanding blood volume, demands of fetus & placenta.
- Vegetarians & blood donors (frequently).

Symptoms of Iron Deficiency Anemia (IDA)

- Clinical symptoms of IDA are non-specific, hence must be diagnosed using laboratory tests. These includes:

- ✓ Extreme fatigue & weakness
- ✓ Pallor (pale skin)
- ✓ Chest pain, fast heartbeat or shortness of breath
- ✓ Headache, dizziness (confused & unable to think clearly) or lightheadedness & dimness of vision.

- ✓ Cold hands and feet
- ✓ Spoon shaped & brittle nails
- ✓ Inflammation or soreness of tongue
- ✓ Unusual cravings for non-nutritive substances, such as ice, clay or starch.
- ✓ Poor appetite, especially in infants and children.

Indicators of the Iron status of populations

Category I: Red blood cell parameters: use **whole blood** as a sample.

1. Haemoglobin (Hgb)
2. Hematocrit (HCT) or Packed Cell Volume (PCV)
3. Mean Cell Volume (MCV) & Mean Cell Hemoglobin (MCH)
4. Red cell Distribution Width (RDW)
5. Reticulocyte haemoglobin concentration and percentage of hypochromic red cells

Category II: Index of size of iron stores:

1. Bone marrow Biopsy
2. Zinc protoporphyrin or Erythrocyte protoporphyrin
3. Serum or Plasma Iron
 - 3.1 Total Iron-binding Capacity
 - 3.2 Percentage of Serum Transferrin Saturation
4. Serum Ferritin
5. Serum Transferrin Receptor

Category I: Red blood cell Parameters

1. Haemoglobin (Hgb):

- Measurement of Hgb is the most widely used test for iron deficiency anemia (IDA).
- Haemoglobin concentration is a measure of **anaemia** (a condition in w/c hemoglobin content of the blood is lower than normal).
- Method/ equipment used: **HemoCue photometer**.

WHO (2011) [Hgb] cutoff points to define Anemia

Age Group	Hemoglobin Concentration, g/dL	Hemoglobin Concentration, g/l
Children aged 6–59 mo	<11.0	<110
Children aged 5–11 y	<11.5	<115
Children aged 12–14 y	<12.0	<120
Adult males (>15 y)	<13.0	<130
Adult females, nonpregnant (>15 y)	<12.0	<120
Adult females, pregnant (1 st & 2 nd trimester)	<11.0	<110
PW (3 rd trimester)	<10.5	<105

WHO Grading of Anemia:

- Grade 1 (Mild Anemia): 10 g/dl
- Grade 2 (Moderate Anemia): 7-10 g/dl
- Grade 3 (Severe Anemia): below 7 g/dl

Factors to be considered during assessment of Iron using Hemoglobin

1. **Pregnancy:** expansion of both plasma volume & red cell mass.
 - The greater expansion of plasma volume, the more Hgb becomes diluted w/c results in a fall in Hgb concentration.
 - ✓ Evident near the end of 2nd trimester of pregnancy.
 - ✓ During the 3rd trimester, Hgb concentration gradually rises.
2. **Deficiencies other micronutrients** such as; vitamin A, C, B6, B12, B2, folic acid, and copper are associated with anemia.
3. **Higher Altitudes:** Hgb concentrations increasing gradually with altitude (above 1000 meters).
 - ✓ Body's adaptive response to lower partial pressure of oxygen leads to reduced oxygen saturation of blood.
4. **Parasitic infections:** Malarial protozoan (*Plasmodium falciparum*) & Helminthes such as hookworms & flukes (schistosomiasis).
Cause low Hgb concentrations by rupturing erythrocytes & suppressing the production of new erythrocytes.
5. **Certain disease states:** Chronic infections & inflammation, chronic disease states including HIV/AIDS, hemorrhage, PEM etc can lower Hgb concentrations. In contrast, polycythemia (increase in RBCs) and hemo-concentration caused by dehydration elevates Hgb values.
6. **Cigarette smoking:** Associated with higher concentrations of Hgb (3 -7 g/L) in adults, due to a reduction of the oxygen carrying capacity of blood arising from carbon monoxide induced increase in carboxyhemoglobin levels.

Summary of Indicators of the Iron status of populations (Category I: Red blood cell Parameters)

Name of Indicator	Indicator of	Sample	Commonly used Method	Notes
Haemoglobin (Hgb)	Anaemia	Whole blood	HemoCue photometer	alone has low specificity & sensitivity
Hematocrit (HCT) or Packed Cell Volume (PCV)	Proportional volume of RBCs in whole blood		Centrifugation of whole blood in capillary tube	After centrifugation, the height of the red cell column is measured and compared to the total height of the column of whole blood. Need stable power supply and well standardized equipment.
Mean Cell Volume (MCV) & Mean Cell Hemoglobin (MCH)	MCV: average RBC size: (microcytic); or (macrocytic) MCH: Hgb in average RBC; (hypochromic, normochromic)		Hemocytometer, particle counter e.g. Coulter counter	Most widely used value for evaluation of nutritional anemia but requires high cost & technical support. A low MCV may also encountered in Thalassemia.
Red cell Distribution Width (RDW)	Abnormal range in size of RBCs <11.5% or >14.5%.		Automated flow cytometry $RDW = SD \text{ of } MCV / MCV$	Increases in iron deficiency anemia. Nonspecific & has high values in B-12 & folate deficiency
Reticulocyte Hgb [C] & percentage of hypochromic red cells	Concentration of hemoglobin in new RBCs.		Automated flow cytometry	Reliable method & has been compared to the “gold standard”. Requires a specific brand of flow cytometer & i.e. currently not helpful in developing countries. Recommended for the evaluation of the adequacy of the iron supply in patients undergoing dialysis for renal failure.

Summary of Indicators of the Iron status of populations (Category II: Index of size of iron stores & iron in transit)

Name of Indicator	Indicator of	Sample	Commonly used Method	Notes
Bone marrow Biopsy	Depleted or absent body iron stores.	Bone marrow aspirate	Microscopical examination of stained marrow cells.	Gold standard for diagnosing iron deficiency using Perls' stain, but expensive, uncomfortable, require technical expertise, and are not performed routinely in clinical practice. The aspirate is spread on to glass slides, air dried, & fixed with methanol, then stained in 10 g/l potassium ferrocyanide for 10 minutes & washed in tap water for 20 minutes and counterstained in 0.1% aqueous safranin.
Zinc protoporphyrin or Erythrocyte protoporphyrin	Insufficient iron in the bone marrow. (Normal EP [C] $\leq 40\text{--}50$ $\mu\text{g/dl}$, in severe iron deficit raises to 200 $\mu\text{g/dl}$).	Whole blood	The ratio of ZnPP to haem is measured by Hematofluorometer / Hemato-fluorimetry method	Has a high sensitivity in diagnosing ID in adults. The last step in hemoglobin synthesis is the insertion of iron by the enzyme ferrochetalase, but a lack of iron results incorporation of trace but measurable amounts of zinc into protoporphyrin w/c leads to increased ZnPP. ZnPP can be increased by lead poisoning, chronic infections, inflammation & hemolytic anemia (RBC anemia) etc.
Serum or Plasma Iron	Iron bound to transferrin in blood.	Serum or plasma	Colorimetry	The plasma or serum iron is the fraction of all iron in the body that circulates bound primarily to transferrin. Normal plasma iron concentration ranges from 50–120 $\mu\text{g/dl}$, iron deficiency range from 50–60 $\mu\text{g/dl}$. Plasma iron concentration can be altered by infections & inflammations.

Summary of Indicators of the Iron status (Index of size of iron stores & iron in transit) cont'd...

Name of Indicator	Indicator of	Sample	Commonly used Method	Notes
Total Iron-Binding Capacity (TIBC)	Total capacity of circulating transferrin bound to iron.	Serum or plasma	Colorimetric assay	TIBC is more stable indicator of iron status, but TIBC by itself is not often used since it doesn't change until iron stores are depleted.
Serum Transferrin Saturation	Saturation of <15% with high TIBC indicates iron deficiency.	Serum or plasma	Enzyme-linked immunosorbent assay (ELISA) or immunoblotting	Widely used screening test for ID, calculated as the ratio of plasma Fe to total Fe-binding capacity. The concentration of transferrin increases during iron deficiency and decreases with protein deficiency. Relatively inexpensive, but its use is limited by diurnal variation in serum iron & many clinical disorders.
Serum Transferrin Receptor (TfR)	Concentration serum TfR raises in IDA <i>i.e.</i> a good indicator of tissue iron deficiency.	Serum or Plasma	Immunoassay <i>e.g.</i> ELISA	TfR is a glycoprotein that transfers circulating iron in to developing red cells, <i>i.e.</i> regulates uptake of transferrin iron into all body cells. It is a marker of the severity of iron insufficiency only when iron stores have been exhausted. Mean serum TfR concentrations tend to be in the range of 5–8 mg/l in normal subjects but can be much higher (25 mg/l), during iron deficiency. Growth, diseases (including thalassemia, megaloblastic anaemia or hemolysis due to malaria), may increase erythropoiesis and TfR independent of iron status. Thus, its value for IDA is uncertain in infections & malaria endemic areas.

Summary of Indicators of the Iron status of populations (Index of size of iron stores & iron in transit) cont'd...

Name of Indicator	Indicator of	Sample	Commonly used Method	Notes
Serum Ferritin (SF)	Size of total body iron stores.	Serum or plasma	Immunoassay <i>e.g.</i> enzyme-linked immunosorbent assay (ELISA) or immunoturbidometry.	Iron stores exist primarily in the form of ferritin (iron-binding protein found in the liver). Ferritin no longer reflects the size of the iron store in the body during inflammation (it's concentration increases since it's an acute phase protein). SF concentration of <15 µg/l reflects iron deficiency, and <12 µg/l in children <5 years old & <30 µg/l in the presence of infection indicates depleted iron stores (WHO). In healthy individuals, it is directly proportional to iron stores: 1mg/l SF corresponds to 8–10 mg body iron. Serum ferritin may be the most useful, widely available & well-standardized method for identifying iron status. But unreliable in the setting of malignancy (cancerous growth), hyperthyroidism, liver disease, and heavy alcohol intake, & during pregnancy, as concentrations fall during late pregnancy, even when bone marrow iron is present.

Assessment Methods of Vitamin A status

- Vitamin A deficiency (VAD) is an important public health problem among children & women in developing countries.
 - Night blindness >5% among women's of child bearing age/ pregnant women (moderate public health problem).
 - Night Blindness >1% & Bitot's spot >0.5% among children 2–5 years old, (severe public health problem) and
 - Serum/ plasma retinol (<0.70 $\mu\text{mol/L}$ or 20 $\mu\text{g/dL}$) in pre-school children: >10 - < 20% (moderate) & $\geq 20\%$ is considered as severe public health problem (WHO, 2010).
- Supplementation, food fortification, nutrition education, and home gardening projects are some of the strategies being used to address this problem.

Biochemical Indicators of Vitamin A status

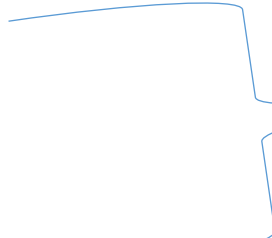
1. Qualitative and Quantitative Biochemical indicators

1.1. Serum Retinol concentration

1.2. Breast milk Retinol concentration

1.3. Retinol Binding Protein concentration

Qualitative indicators



1.4. Vitamin A store in the liver: Relative Dose Response (RDR) & Modified Relative Dose Response (MRDR): Quantitative indicators.

2. Biological, Functional and Histological indicators

1. Qualitative biochemical indicators of vitamin A status

1.1. Serum Retinol Concentration

- Frequently used biochemical indicator of vitamin A status.
- Serum retinol concentration reflects vitamin A status best, if the person has exhausted stores of the vitamin in the liver.
 - Because serum retinol concentration is homeostatically controlled and will not drop until body stores are significantly compromised.
- Use of serum retinol is **unreliable** as an indicator of vitamin A status in people with **infection** or **inflammation**, **poor protein status** and **organ diseases**.

Serum Retinol Concentrations... (1)

- Serum retinol is usually assessed by **high-performance liquid chromatography (HPLC)** or **spectrophotometry**.
 - ✓ HPLC can detect a concentration of $<0.70 \mu\text{mol/L}$ with adequate precision (but not spectrophotometric analysis).
- Serum retinol concentration of $0.70 \mu\text{mol/L}$ is a well-recognized cut-off.
 - ✓ Less than $10 \mu\text{g/dL}$ ($0.35 \mu\text{mol/L}$): indicates deficiency.
 - ✓ $10\text{--}20 \mu\text{g/dL}$ [$0.35\text{--}0.7 \mu\text{mol/L}$]: suggests marginal status & inadequate stores.
 - ✓ $30\text{--}86 \mu\text{g/dL}$ ($1.05\text{--}3 \mu\text{mol/L}$): adequate vitamin A status.
 - ✓ Greater than $86 \mu\text{g/dL}$ ($>3 \mu\text{mol/L}$): excess (hypervitaminosis A).

Serum Retinol Concentrations ...(2)

Factors affecting Serum Retinol levels

- **Age**: serum retinol levels increase with age.
- **Sex**: adult males have higher values of serum retinol than females.
- **Low fat diets**: ($\sim < 5$ to 10 g daily).
- **Impair absorption of provitamin A carotenoids**.
- **Zinc deficiency**: lowers serum retinol, impairs synthesis of RBP.
- **Infection/ inflammation**: lower serum retinol.
- **Liver vitamin A stores** 0.7-1.05 $\mu\text{mol/g}$: serum retinol is homeostatically controlled, $< 0.7 \mu\text{mol/g}$: serum retinol falls; & $> 1.05 \mu\text{mol/g}$: serum retinol rises.

1.2. Breast Milk Retinol concentration

- Vitamin A is found in the fat portion of breast milk.
- Breast milk retinol concentration can be used as evaluation tool of both maternal and infant vitamin A status.
- It reflects recent dietary intake *i.e.* provides information on the risk of inadequate VA intake.
- It can be measured by HPLC or fluorometry.
- A cut off value of:
 - ✓ Less than or equal to 8 $\mu\text{g/g}$ milk fat or $\leq 1.05 \mu\text{mol/L}$ suggested as vitamin A deficient.
 - ✓ Below 10 $\mu\text{g/g}$ indicates that the infant is at a risk of VAD.
 - ✓ Above 40 $\mu\text{g/g}$ are considered as adequate.

1.3. Retinol-Binding Protein (RBP) Concentrations

- RBP is a carrier protein for vitamin A and binds one molecule of retinol.
- Serum RBP concentration decreased in vitamin A and zinc deficiencies, (zinc is required for RBP synthesis from the liver).
- Vitamin A and/or zinc supplementation, increases serum RBP level.
- The ratio of retinol to RBP is influenced by vitamin A deficiency.
- During VAD, RBP accumulates in the liver and released unbound to retinol.

Quantitative Biochemical indicators:

Vitamin A stores in the liver

- Most of the Vitamin A is stored in liver in the form of retinyl ester.
- Thus, a measure of liver vitamin A stores is the best index.
- VAD: “when liver stores of retinol are below 20 μ g”.
- Adequacy of liver vitamin A stores can be assessed by quantitative biochemical tests including;
 - i. The Relative Dose Response (RDR) test
 - ii. The Modified Relative Dose Response (MRDR) test

i. The Relative Dose Response (RDR) Test

- RDR test is more descriptive than serum retinol concentrations but invasive (requires taking 2 blood samples from the same individual within 5hrs).
- RDR work on the principle that, as liver reserves of vitamin A become low, RBP accumulates.
- When 1mg of retinyl ester is administered orally, it binds to the accumulated RBP and is rapidly released into the serum.
- RDR test involves measuring changes in plasma retinol concentration before (blood is taken initially), and 5 hours after administration of retinyl esters (blood is taken again).

$$\text{RDR \%} = \frac{\text{5-hour plasma retinol} - \text{initial plasma retinol}}{\text{5-hour plasma retinol concentration}} \times 100$$

- **RDR $\geq 20\%$ suggests inadequate liver vitamin A ($< 0.7 \mu\text{mol/g}$).**

ii. The Modified Relative Dose Response (MRDR) Test

- The test works on the same principle as the RDR test.
- Involves measuring the ratio of 3,4 didehydroretinol (vitamin A2) to retinol (vitamin A1) in the blood after administering a single dose of 3,4 didehydroretinyl acetate orally instead of retinol.

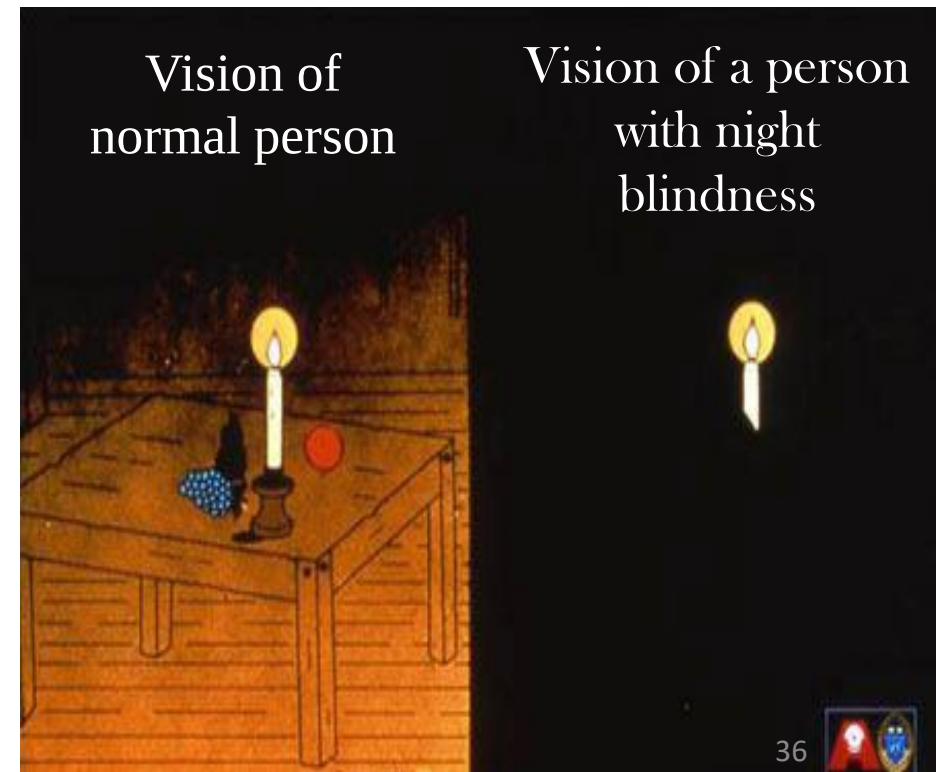
$$\text{MRDR ratio} = \frac{\text{Plasma vitamin A2 concentration}}{\text{Plasma vitamin A1 concentration}}$$

- HPLC easily separates 3,4-didehydroretinol from retinol.
- This test, unlike the RDR, requires taking only one blood sample, about 4 to 7 hours after the vitamin is ingested.
- A cut-off of 3,4-didehydroretinol to retinol ≥ 0.06 suggests low liver reserve of retinol that can be corrected with vitamin A supplementation.

2. Biological, Functional and Histological Indicators

2.1. Night blindness:

- The first detectable functional or clinical sign of VAD.
- Manifested by: inability to see in dim light or inability to recover sight after a flash of bright light.
- Occurs when the vitamin A pool in the eye becomes depleted & the concentration in the rod cells is lowered.
- May occur in **marginal vitamin A status** & **during pregnancy**.



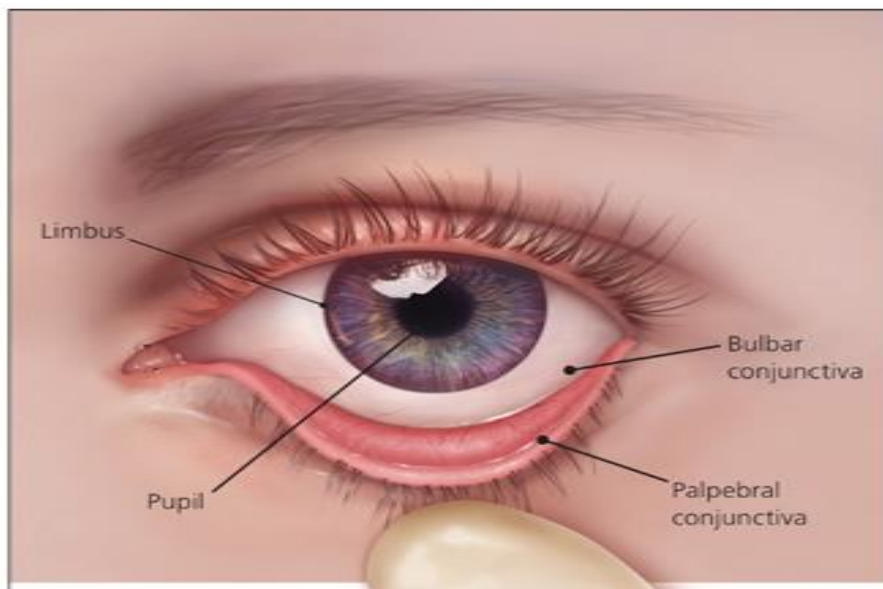
Night Blindness... Cont'd

- Night blindness due to vitamin A deficiency is reversible with increased vitamin A intake or supplementation.
- If left untreated the night blindness is followed by **xerophthalmia**.
- It can be assessed by:
 - **Electrophysiological technique (Electroretinograms)**.
 - ✓ Measures the level of rhodopsin and its rate of regeneration.
 - **Vision Restoration time**:
 - ✓ An objective assessment technique, that measures the ability of the bleached eye to recognize a letter under dim light and thus can assess dark adaptation.

Functional indicators... (1)

2.2. Xerophthalmia: extreme dryness of the conjunctiva.

- A disease associated with structural changes in the cornea (front cover of the eye).
- The first visible structural change is:
 - ✓ Drying of the conjunctiva and the cornea (**xerosis**) and
 - ✓ The development of white foamy patch on the conjunctiva called **Bitot's spot** (build-up of keratin).
- This is followed by the development of **keratomalacia**: (softening, deterioration & ulceration of the cornea).
 - Causes irreversible damage to the cornea and leads to **blindness**.



scarring of the cornea.

Image: The dryness seen here is known as xerophthalmia. White foamy patch on the conjunctiva called **Bitot's spot**.

Functional indicators... (2)

2.3. Keratinization of Epithelial cells

- In addition to keratinization of the cornea, VAD results in keratinization of epithelial cells.
- Epithelial cells secrete a protein called keratin.
- Changes in epithelial cells results in keratinization, rough, dry & scaly skin.



Functional indicators... (3)

2.4. Retarded growth

- Xerophthalmia was reported to be:
 - Accompanied by upper respiratory infection and diarrhea.
 - Exacerbated by protein energy malnutrition.
 - Increased susceptibility to infection (lower resistance to infections) and with increased mortality in young children.
 - In general retarded growth.
- Deficiency disease is called hypovitaminosis A.

Other Biochemical Indicators Vitamin A status

■ Ecological & Socio-demographic indicators

- Illiterate women 15–44 years of age, households those spend more than 50% of income on food, have no safe water supply, poor villages over 10km far from health services, inadequate access to land/agriculture inputs, urban households with no regular income.

■ Nutritional status and diet related indicators:

- Breastfeeding patterns, in availability of vitamin A rich foods at home and in the market,
- Prevalence of low birth weight (LBW), stunting & wasting among children <5 years of age.

■ Illness related indicators:

- Measles, diarrheal disease, fever and helminthic infection are useful in identifying populations at risk of vitamin A deficiency.

Assessment of Zinc Status

- Zinc is an essential mineral for human health.
- Abundant in nature & all human organs, tissues, fluids, & secretions, and >95% in fat-free mass, mainly in intracellular compartments.
- **More than 300 enzymes** require Zinc for function or regulation.
 - ✓ Synthesis of nucleic acids, amino acids & proteins.
 - ✓ Wound healing, growth & reproduction,
 - ✓ Interactions with hormones: insulin & adrenal corticosteroids.
 - ✓ Vitamin A metabolism.
 - ✓ Appetite & taste acuity.
 - ✓ Immune function &
 - ✓ Neuro-behavioural function.

Causes of Zinc deficiency

- Zinc deficiency is the most important causes of morbidity in developing-countries.
- Causes of zinc deficiency includes:
 - Inadequate dietary zinc intakes.
 - Consumption of diets high in **phytates** (cereals & legumes), **high dietary phosphate**, **excessive calcium & dairy products**, **high fibre** and **coffee** can decrease Zinc absorption.
 - **High physiological requirements**: infancy, childhood, adolescence, pregnancy & lactation.
 - **Excessive losses**: strenuous exercise; sweat & losses in urine (Alcoholism), diarrhea etc.
 - Diabetes mellitus,
 - **Drugs**; Thiazide diuretics (for HBP): increase urine output, penicillamine (use: removal of toxic metals from the body) etc. 44

Health Consequences of Zinc Deficiency

▪ Infancy

- ✓ Delayed immune system development: increased susceptibility to infections & diarrhoea.
- ✓ Mood changes: irritability.
- ✓ Impaired cognitive function.
- ✓ Poor appetite, impaired taste acuity.
- ✓ Failure to thrive (grow healthily).

▪ Adolescence

- ✓ Slower skeletal growth (growth retardation).
- ✓ Reduced bone mineralization.
- ✓ Disturbances in body composition.
- ✓ Delayed secondary sexual maturation.

▪ Childhood

- ✓ Poor appetite.
- ✓ Disturbances in body composition & impaired linear growth.
- ✓ Increased susceptibility to infections: diarrhoea & Pneumonia.
- ✓ Increased mortality.

▪ Pregnancy

- ✓ Adverse maternal outcomes: **abortion, miscarriage, stillbirths, prolonged labour & postpartum hemorrhage, preterm delivery, LBW, congenital malformations etc.**
- ✓ **Pregnancy-induced hypertension & impaired glucose tolerance,**

Assessment of risk of Zinc status in populations

- Zinc deficiency usually leads to very general signs or symptoms such as: growth impairment & general immune impairment.
 - ✓ These symptoms can also result from many other causes.
 - ✓ No specific syndrome specifically point to zinc deficiency.
- As a result, there is lack of clear indicator to assess zinc deficiency.
- The following indicators can be used to assess zinc status:
 1. Serum (or Plasma) Zinc concentration
 2. Hair Zinc concentration
 3. Urinary Zinc concentration
 4. Functional indicators
 5. Indirect indicators: rate of stunting & rate of anemia.

Assessment of the risk of Zinc status... (1)

1. Serum (or Plasma) Zinc concentration

- The most widely used biomarker to determine zinc status.
- It reflects dietary zinc intake/ bioavailability of zinc.
- **Low serum zinc concentrations:** indicate possible **zinc deficiency**.
- *NB: Plasma zinc turns over 150 times a day in normal individual.*
- Cut-offs for zinc deficiency using serum zinc concentrations:
 - ✓ **Less than 9.9 $\mu\text{mol/L}$** (morning) or **<8.7 $\mu\text{mol/L}$** (afternoon) for non-fasting children <10 years.
 - ✓ **Less than 10.1 $\mu\text{mol/L}$** (morning) and **<9.0 $\mu\text{mol/L}$** (afternoon) for non-fasting women of reproductive age (aged 18–49 years).
 - ✓ Normal reference range **10-17 $\mu\text{mol/L}$** (in plasma).
 - ✓ Public health concern: prevalence of low serum zinc **>20%**.

Serum (or Plasma) Zinc concentration... (1)

- Deprivation of zinc results in a reduced plasma zinc concentrations.
- However, plasma zinc concentrations are affected by many other factors, including:
 - ✓ **Inflammation** (causes depression of zinc concentration),
 - ✓ **Fasting** (leads to higher zinc concentration) or **eating** (results in a decrease in serum zinc concentrations),
 - ✓ **Pregnancy** (plasma zinc levels lower in late pregnancy),
 - ✓ **Use oral contraceptive** (negatively affect Zinc level),
 - ✓ **Diurnal rhythm** (as large as 20%),
 - ✓ **Age & sex**: lower in childhood, peaked during adolescence & young adulthood, declined slightly thereafter, and higher average contents of zinc in men than in women.

Assessment of the risk of Zinc status... (2)

2. Hair Zinc concentration

- Hair, skin, & nails consists of 20% of body zinc .
- Low hair zinc concentration is index of longer-term (chronic) zinc deficiency.
- Hair zinc concentrations differ markedly according to sex & seasons.
 - Boys have lower hair zinc concentrations than girls of the same age, even when food consumption patterns, energy & nutrient intakes are comparable.
 - ✓ Because higher zinc requirements for boys or changes in growth hormone & testosterone.
 - ✓ Hair zinc values in infants and young children of $<70 \mu\text{g/g}$ ($1.07 \mu\text{mol/g}$) in the spring or summer and $<110 \mu\text{g/g}$ ($1.68 \mu\text{mol/g}$) in the winter: Mild zinc deficiency.

Assessment of the risk of Zinc status... (3)

3. Urinary Zinc Concentrations

- In normal circumstances, zinc excreted with urine accounts for around 15% of the daily losses.
- However, when consuming a diet low in zinc, the amount of zinc excreted through urine is reduced by 96%.
- Zinc intake is related to urinary zinc concentrations; in subjects with a **moderate zinc status**.
- Currently there is not enough data on urinary zinc to make solid recommendations regarding its validity & usefulness as indicator for zinc status.

Assessment of the risk of Zinc status... (4)

4. Functional indicators of Zinc status

- Immune function
- Rates of infection: diarrhea & respiratory tract infections.
- Cognitive function
- Appetite

Assessment of the risk of Zinc status... (5)

5. Indirect indicators: Imprecise and may lead to erroneous estimates of zinc deficiency.

5.1. Rate of Stunting

- Zinc supplementation has a positive effect on length growth, in children under 2 years of age & stunted children.
- Zinc deficiency is associated with growth faltering and stunting.
- However, growth faltering can be occurred due to many factors (including inadequate intake of energy and/or protein, & deficiency of other nutrients).
- Mainly applied for children under the age of 5 years.

Indirect indicators of Zinc status... cont'd

5.2. Rates of Anemia (Anemia Prevalence)

- Iron and zinc have a similar distribution in the food supply and their absorption is inhibited by phytates.
- Meat & other animal flesh foods are richest sources of iron & zinc, and cereal grains have moderate content of iron & zinc.
- The nutritional causes of iron and zinc deficiencies are similar.
- So, high rates of iron-deficiency anemia may be a suggestive evidence of the risk of zinc deficiency.

Conclusion

- Toddlers, pregnant women & women of child bearing age (specially who exclude flesh foods) are among sub-population groups at risk of Zinc deficiency.
- Zn deficiency is often dietary-induced.
- Meat group is the major source of readily available Zinc.
- Vegetarianism have a negative impact on bioavailability of dietary zinc.

Assessment of Iodine Status

Sources and Physiological Functions of Iodine

- Iodine, is an essential component of the thyroid hormones; namely: **Triiodothyronine (T3)** and **Thyroxin (T4)**.
- The human body contains 15–20 mg of iodine, of which 70–80% is concentrated in the thyroid gland.
- Iodine is obtained from **diet & a component of some medications**.
- T3 & T4 are involved in regulating the body's metabolic processes by controlling energy production & oxygen consumption in cells for normal growth and development, neural and sexual development.
- When iodine intake falls below recommended levels, thyroid may not be able to synthesize sufficient amounts of thyroid hormones.

Iodine deficiency situations

- Low level of thyroid hormones in the blood (hypothyroidism) affects the development & functioning of brain, muscles, heart, liver, and kidneys.
 - These results in a range of poor health conditions collectively known as **iodine deficiency disorders (IDDs)**.
- Globally, about 30% of the world's populations affected with IDD.
- In Ethiopia, IDD is a major public health problem.
- The prevalence of goiters in children 6 -12 years of age was 40%.
- Nearly 46% of the children had median urinary Iodine excretion levels below 20 $\mu\text{g/l}$ indicating moderate to severe IDD.

Iodine intake Recommendations

- ICCIDD, UNICEF and WHO recommends the following daily intake of iodine:
 - For preschool children (0 to 59 months): **90 µg**.
 - For school children (6 to 12 years): **120 µg**.
 - For adolescents (above 12 years) and adults: **150 µg**.
 - For **pregnant & lactating women**: **250 µg**.
- Iodine requirements are higher in pregnancy because of:
 - ✓ Increased thyroid hormone production.
 - ✓ Increased renal iodine excretion, and
 - ✓ Fetal iodine requirements.

Causes of IDD's

- Soil devoid of iodine.
- Poor consumption of sea foods.
- Increased consumption of foods of plant origin (cabbage, broccoli & cassava) contain goitrogens.
- Co-existing deficiency of other micronutrients (Fe, Se, Vitamin A).
 - ✓ Iron deficiency- reduces the iron-dependent hemo-protein thyroperoxidase enzyme catalyze.
 - ✓ Selenium deficiency- impaired production of the selenium-dependent glutathione peroxidase.
 - ✓ Deficiency of vitamin A- activates thyroid stimulating hormone (TSH) and increases risk of goiter.

Table: The spectrum of iodine deficiency disorders (IDDs)

PHYSIOLOGICAL GROUPS	HEALTH CONSEQUENCES OF IODINE DEFICIENCY
All ages	Goitre Hypothyroidism Increased susceptibility to nuclear radiation
Fetus	Spontaneous abortion Stillbirth Congenital anomalies Perinatal mortality
Neonate	Endemic cretinism including mental deficiency with a mixture of mutism, spastic diplegia, squint, hypothyroidism and short stature Infant mortality
Child and adolescent	Impaired mental function Delayed physical development Iodine-induced hyperthyroidism (IIH)
Adults	Impaired mental function Iodine-induced hyperthyroidism (IIH)

^a Adapted from BS Hetzel, 1983 (7).

Target groups for Assessment of Iodine status

- The preferred target group is **school age children between 6 and 12 years**, preferably, 8-10 years old children.
 - ✓ Because, they are **vulnerable to iodine deficiency** & **accessible**.
- **Pregnant women** are a prime target group for **IDDs** control activities.
 - ✓ Because they are especially **sensitive to marginal iodine deficiency** and relatively **accessible** (in antenatal clinics).

Assessment Methods Iodine status

- Iodine status can be assessed using the following methods.
 1. Urinary Iodine Excretion/ concentration (UIE)
 2. Thyroid Size
 3. Serum Thyroglobulin (Tg)
 4. Thyroid Stimulating Hormone (TSH)
 5. Serum Triiodothyronine (T3) or Thyroxine (T4) concentrations
 6. Functional indicators of Iodine status

Assessment Methods Iodine status... (1)

1. Urinary Iodine Excretion/ concentration (UIE)

- Most of the iodine absorbed in the body appears in the urine.
- Therefore, UIE is a good marker & more sensitive indicator of **very recent dietary iodine intake**.
- UIE can vary somewhat from day to day & even within a given day.
- Two main ways to collect urine for iodine measurement are:
 - i. Collection of urine over a period**, usually **24 hours** ($\mu\text{g/L}$, or μg iodine/g creatinine): often impractical to obtain.
 - ii. Collection of urinary spot iodine sample**: high day-to-day variability in iodine intake & in water consumption.

Criteria for assessing iodine nutrition based on median urinary iodine concentrations (UIC) of school-age children (≥ 6 years)*

Median UIC ($\mu\text{g/L}$)	Iodine intake	Iodine status
< 20	Insufficient	Severe iodine deficiency
20-49	Insufficient	Moderate iodine deficiency
50-99	Insufficient	Mild iodine deficiency
100 -199	Adequate	Ideal iodine nutrition
200 -299	Above requirement	Likely to provide adequate intake for PLW : and more than adequate intake in over all population.
≥ 300	Excessive	Risk of adverse health consequences (iodine-induced hyperthyroidism & autoimmune thyroid diseases).

* Applied to adults, but not to pregnant & lactating women (PLW).

Criteria for assessing iodine nutrition based on median urinary iodine concentrations for pregnant & lactating women (PLW)

Median urinary iodine ($\mu\text{g/l}$)	Iodine intake
---	---------------

Pregnant women

< 150	Insufficient
150–249	Adequate
250–499	More than adequate
≥ 500	Excessive*

Lactating women†

< 100	Insufficient
≥ 100	Adequate

Children less than 2 years old

< 100	Insufficient
≥ 100	Adequate

* The term 'excessive' means in excess of the amount required to prevent and control iodine deficiency.

† In lactating women, the figures for median urinary iodine are lower than the iodine requirements because of the iodine excreted in breast milk.

Assessment Methods Iodine status... (2)

2. Thyroid Size

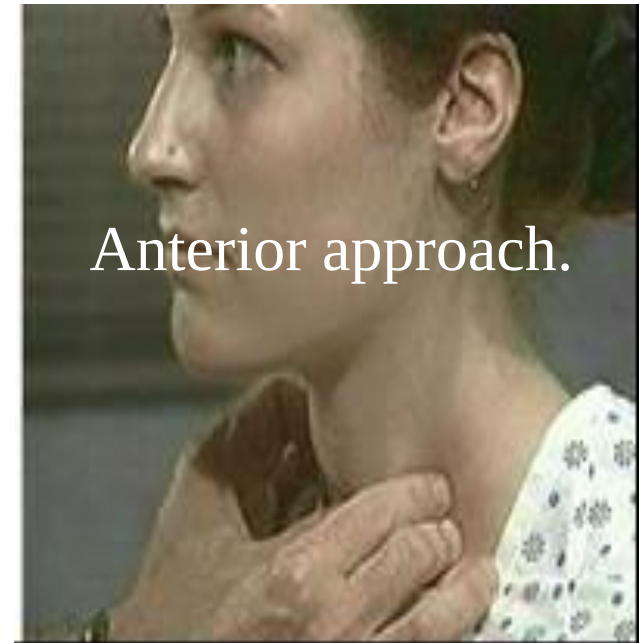
- The size of thyroid gland changes inversely in response to alterations in iodine intake.
- Thyroid size can be determined by neck inspection and palpation.
- This method enables to examine a large number of people in a short time without using expensive equipment.
 - However, it requires careful training and experienced observer.
- A thyroid gland each of whose lobes have a volume greater than the terminal phalanges of the thumb of the person examined will be considered **goitrous**.
 - Endemic goiter: if $\geq 10\%$ of child population has goiter in an area.
- Thyroid size reflects iodine status over a period of months or years.

2.1. Thyroid size by neck Inspection

- Can be done by anterior & lateral approaches.
- The subject seated or standing in a comfortable position with the neck in a neutral or slightly extended position (to enhance visualization of the thyroid).
- Have the subject swallow a sip of water, watching for the upward movement of the thyroid gland.
- After completing anterior inspection of the thyroid, observe the neck from the side (lateral approach).

2.2. Thyroid size by Palpation

- Can be done in anterior or posterior approach.
- The subject is examined in the seated or standing position.
- Have the subject swallow a sip of water as you palpate, feeling for the upward movement of the thyroid gland.



Grade 0	No palpable or visible goiter
Grade 1	An enlarged thyroid <i>i.e.</i> palpable but not visible w/n the neck is in a normal position. It moves upward when the subject swallows.
Grade 2	Clearly visible swelling when the neck is in normal position & is consistent with an enlarged thyroid when palpated.

2.2.1. Total Goiter Rate (TGR)

- TGR is calculated as the sum of grades 1 and 2.
- When this rate **exceeds 5%** in school children (6–12 years old) the population is said to have a **public health problem**.
- Changes in the goiter rate reflect long-term iodine nutrition (months to years).
- Usually IDD is corrected rapidly with effective salt iodization program but goiter rates take longer to return to an acceptable level.

Severity of IDD	Prevalence of Goiter (TGR %)
Iodine sufficiency	<5 %
Mild IDD	5-19.9%
Moderate IDD	20-29.9%
Severe IDD	≥30%

2.3. Thyroid size by Ultrasonography

- Ultrasonography provides a more accurate and objective method of determining thyroid size than palpation.
- It is a safe, non invasive, quickly done (2-3 minutes/subject), using specialized technique and an expensive equipment.
- It is useful in distinguishing between goiter grades 0 and 1.



Assessment Methods Iodine status... (3)

3. Serum Thyroglobulin (Tg)

- Thyroid specific, most abundant intrathyroid protein.
- A key precursor in the production of thyroid hormones.
- In **iodine sufficiency**, small amounts of Tg are secreted into the circulation, and **serum Tg is normally $<10 \mu\text{g/L}$** .
- In areas with **IDDs (endemic goiter)**, **serum Tg increases** due to greater thyroid cell mass and TSH stimulation.
- *NB*: Serum thyroglobulin reflect an intermediate response to iodine status (weeks to months).

Assessment Methods Iodine status... (4)

4. Thyroid Stimulating Hormone (TSH) or Thyrotropin

- TSH is released from the pituitary gland and regulates the synthesis and release of thyroid hormones.
- Serum TSH concentration increases in iodine deficiency.
- Measurement of TSH in serum is used as a screening test for detecting congenital hypothyroidism in neonates.
- Reason: because the neonatal thyroid has limited iodine stores and even mild iodine deficiency may increase TSH secretion.

Assessment Methods Iodine status... (4)

5. Triiodothyronine (T3) or Thyroxine (T4) concentrations

- T3 & T4 levels are regulated by thyroid stimulating hormone (TSH).
- Levels of T3 or T4 in serum can be used to assess iodine status.
- But they are insensitive or poor indicators, because;
 - ✓ In iodine deficient populations serum T3 may remain unchanged & serum T4 usually decreases but the change is within the normal range (overlap with iodine sufficient populations).
- Low circulating T4 causes an increase in TSH levels & this increases the amount of T4 released into circulation.
- Low T3 & T4 levels with elevated TSH indicates hypothyroidism, where as high T3 & T4 levels with low TSH indicate hyperthyroidism.
 - Total T4 = 5.0-12 µg/dL, Free T4 = 1.0-3.0 ng/dL.
 - Total T3 = 80-190 ng/dL, Free T3 = 0.25-0.65 ng/dL.

6. Functional indicators of Iodine status

6.1. Incidence of Cretinism

- Provides an indication of the magnitude of IDD.
- The condition is rare and difficult to diagnose.
- **Cretinism prevalence 0-5% indicates severe iodine deficiency.**

6.2. Cognitive function

- Determination of cognitive function in children has sometimes been used as a functional measure of iodine status.

Conclusion

- Several indices of iodine status are available: choice depends on age or life-stage of subjects, dietary conditions, iodine status of group, and study objectives.
- WHO recommends at least two indicators to reflect IDD.
- Urinary iodine: for population iodine status.
- Serum TSH: for newborns' iodine status.
- Ultrasonography: in areas where iodine deficiency is suspected.
- Iodine intakes above 300 $\mu\text{g/l}$ per day should generally be discouraged, may lead to iodine-induced hyperthyroidism (IIH).
- IIH occurs more commonly in older subjects with pre-existing nodular goiters, and may occur even when iodine intake is within the normal range.

Assessment of Folate and Vitamin B12

Introduction

- Deficiencies of folate or of vitamin B12 are the most common causes of macrocytic anemia.
 - ✓ They affect all age groups and constitute a major global burden of morbidity.

The symptoms of folic acid or vitamin B-12 deficiency include:

✓ Diarrhea or constipation	✓ Bleeding gums	✓ Fatigue & difficulty walking
✓ Lack of appetite	✓ Numbness	✓ Swollen & red tongue
✓ Headaches	✓ Memory loss	

1. Assessment of Folate Status

- Folate & Folic acid are different chemical forms of vitamin B9.
- Folate plays an important role in the human body:
 - ✓ As cofactors in the key metabolic pathways.
 - ✓ Synthesis of DNA, protein & several amino acids, (methionine), metabolism of amino acids (glycine),
 - ✓ Antibody formation,
 - ✓ Normal cell growth & replication,
 - ✓ Normal development of a fetus,
 - ✓ Production of healthy red blood cells etc.

Assessment of Folate Status Cont'd... (1)

- Folate is naturally found in **several foods**, including:
 - ✓ Liver, egg yolks, milk, citrus fruits, whole grains, legumes (beans, lentils & Peas), soybean, fortified cereals, dark green leafy vegetables (spinach & kale), asparagus, broccoli, avocados, Brussels sprouts, etc.
- Naturally occurring folate is less bioavailable than that of folic acid (Tetrahydrofolate: *i.e.* used in supplements & fortified foods).
- **Folate deficiency results from:**
 - Increased renal excretion, malabsorption & impaired folate transport to colon, pancreas & fetus, and ethanol ingestion.
- Folic acid deficiency may also lead to **anemia**.
- Inadequate maternal folate status linked to:
 - Preeclampsia, spontaneous abortion, stillbirth, preterm delivery, low birth weight, and serious congenital anomalies of the brain & spine, such as **neural tube defects**.

Assessment of Folate status Cont'd...(2)

1. Serum, Plasma or Red cell Folate

- Reliable indicator of recent dietary folate intake (*i.e.* fasting concentrations are a better indicator of short-term status).
- Was first carried out using microbiological assays methods (*Lactobacillus casei*) in the 1930s and were the “gold standard” for many years.
- But now a days, they are being replaced by radio isotopic, enzyme-linked, and chemiluminescent assays.
- Serum folate concentrations:
 - ✓ Below 3ng/mL (<7nmol/L): clinical deficiency, negative folate balance, if this persists for weeks or months, folate deficiency becomes evident).
 - ✓ Between 3 and 6 ng/mL marginal deficiency, and
 - ✓ Between 6 and 20 ng/mL normal range.

Assessment of Folate status Cont'd...(3)

2. Competitive Protein Binding Assays

- Widely used in clinical laboratories & population studies.
- Measures the metabolic use of folate.

3. Surrogate Metabolite Assays

3.1 Plasma Homocysteine

- Done by HPLC, GCMS, radioenzymatic assays, and immunoassays.
- Here the plasma should be separated from red cells **as soon as possible following the withdrawal of a blood sample** to avoid *in vitro* exchange between the plasma and cellular compartments or cellular metabolism of homocysteine.

Assessment of Folate status Cont'd...(4)

4. Functional Indicators of Folate Status

- Includes plasma or urinary **formiminoglutamate, aminoimidazole carboxamide riboside, and plasma homocysteine.**
- Elevations of these indicators reflect impaired one-carbon metabolism.
- None of these indicators are specific for folate status since they can also be elevated in **vitamin B₁₂ deficiency** and in the case of homocysteine, **vitamin B₆ & vitamin B₂ deficiency.**

Table 1

Folate concentrations in serum and red blood cells for determining folate status in all age groups, using macrocytic anaemia as a haematological indicator

Serum/plasma folate levels ng/mL (nmol/L) ^{a, b}	Red blood cell folate level ng/mL (nmol/L) ^{a, b}	Interpretation
>20 (>45.3)		Elevated
6–20 (13.5–45.3)		Normal range
3–5.9 (6.8–13.4)		Possible deficiency
<3 (<6.8)	<100 (<226.5)	Deficiency

^aFolic acid conversion factor: 1 ng/mL = 2.265 nmol/L.

^bAssayed by *Lactobacillus casei*.

Table 2

Cut-off values for determining folate deficiency in all age groups using homocysteine concentrations as the metabolic indicator

Folate indicator	Cut-off value indicating folate deficiency ng/mL (nmol/L) ^{a, b}
Serum/plasma folate level	<4 (<10)
Red blood cell folate level	<151 (<340)

^aFolic acid conversion factor: 1 ng/mL = 2.265 nmol/L.

^bMeasured by the radioimmunoassay. In this dataset, the measurements obtained by radioimmunoassay need adjustment to make them comparable with the microbiological assay (10).

Assessment of Vitamin B₁₂ Status

1. Direct Vitamin Assays

1.1 Plasma or Serum Vitamin B₁₂

- Is the most frequently used laboratory test to assess vitamin B₁₂ status in population screening.
- Serum vitamin B₁₂ ~148 pmol/L (200 ng/L) set as the threshold for diagnosing deficiency.
- More recently, Serum B₁₂ assays measure holotranscobalamin B₁₂ (the sum of haptocorrin-bound and plasma binding protein transcobalamin-bound).
- A homocysteine concentration >20μmol/L suggests B₁₂ deficiency in folate-replete patients.

N.B. No single laboratory marker is suitable for the assessment of B₁₂ status in all patients.

Assessment of Vitamin B₁₂ status Cont'd... (1)

2. Surrogate Metabolite Assays

Plasma or Serum Methylmalonate

- Involves a derivatization followed by gas chromatography–mass spectrometry (GCMS).
- Requiring sophisticated equipment.
- Only in **vitamin B₁₂ deficiency**, the levels of methylmalonate in the plasma and urine **rises** (not in Folate deficiency).
- Besides, **impairment of renal function** may also cause an elevation of plasma methylmalonate.

End of Chapter Four

THANKS